塞罕坝

一本日记，三代人，一片人种出来的森林

Saihanba

One diary, three generations, and a forest planted by hand

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Footnotes mark names, festival terms and words above HSK 6 (HSK 3.0 lists: github.com/ivankra/hsk30, MIT).

1 早上五点半，天还没完全亮，赵雨[†]已经站在山顶[‡]上了。九月的塞罕坝[§]，风里已经有了冬天的味道。她打开箱子，取出一架无人机[¶]，看着它慢慢升到空中，越飞越远。她手里的屏幕上出现了一片看不到头的绿色：松树一排一排地站着，像大海上的波浪，时而被风吹弯，时而又站直了。

2 赵雨知道，在这片绿色下面，几乎每一棵树都是人种的。她的背包里放着一个旧本子，纸已经发黄了，第一页上写着一个名字：周玉芬[*]。那是她奶奶的日记，第一篇的日期是一九六二年九月。那一年，这里还几乎看不到一棵树。

† 赵雨 *Zhào Yǔ* Zhao Yu
‡ 山顶 *shāndǐng* hilltop
§ 塞罕坝 *Sàihǎnbà* Saihanba, forest in Hebei
¶ 无人机 *wúrénjī* drone
* 周玉芬 *Zhōu Yùfēn* Zhou Yufen

At half past five in the morning, before it was fully light, Zhao Yu was already standing on the hilltop. In September at Saihanba, the wind already carried a taste of winter. She opened a case, took out a drone and watched it rise slowly into the air and fly farther and farther away. On the screen in her hand appeared a sea of green with no end in sight: row upon row of pines, like waves on the ocean, now bent by the wind, now standing straight again. 1

Zhao Yu knew that beneath this green, almost every single tree had been planted by human hands. In her backpack was an old notebook whose paper had already turned yellow; on the first page was written a name: Zhou Yufen. It was her grandmother's diary, and the first entry was dated September 1962. That year, there was hardly a tree to be seen here. 2

一九六二年的风

3 一九六二年夏天，周玉芬[1]十九岁。她刚从承德[2]的一所林业学校毕业，就报名去了塞罕坝[3]新建的机械林场[4]。塞罕坝在河北省[5]的最北边，离北京的直线距离大约只有一百八十公里。据说，“塞罕[6]”在蒙古语里是“美丽”的意思，“坝[7]”是汉语里的“高地”，合在一起，就是“美丽的高原”。

4 很久以前，这里确实很美。清朝[8]的时候，这里是皇帝打猎[9]的地方，到处是森林、河和草原。后来，树被一片一片地砍[10]掉，又经过几次大火，到了二十世纪五十年代，只剩下黄沙和野草。北边吹来的风沙[11]，没有东西能挡，一直吹到北京。

1 周玉芬 *Zhōu Yùfēn* Zhou Yufen
2 承德 *Chéngdé* Chengde, city in Hebei
3 塞罕坝 *Sàihǎnbà* Saihanba, forest in Hebei
4 林场 *línchǎng* forest farm
5 河北省 *Héběi Shěng* Hebei Province
6 塞罕 *sàihǎn* beautiful (Mongolian)
7 坝 *bà* highland
8 清朝 *Qīngcháo* Qing dynasty
9 打猎 *dǎliè* to hunt
10 砍 *kǎn* to chop
11 风沙 *fēngshā* wind-blown sand

The Wind of 1962

In the summer of 1962, Zhou Yufen was nineteen. She had just graduated 3
from a forestry school in Chengde when she signed up for the newly founded mechanised forest farm at Saihanba. Saihanba lies at the northern tip of Hebei Province, only about 180 kilometres from Beijing in a straight line. It is said that "saihan" means "beautiful" in Mongolian, and "ba" is Chinese for "highland"; put together, the name means "beautiful highland".

Long ago, this place really had been beautiful. In the Qing dynasty it was 4
the emperors' hunting ground, full of forests, rivers and grassland. Later the trees were cut down tract after tract, there were several great fires, and by the 1950s all that was left was yellow sand and wild grass. Wind and sand from the north met nothing to stop them and blew all the way to Beijing.

5 出发以前，同学们跟她开玩笑：“听说坝[1]上的风大得能把人吹走。”周玉芬[2]笑着说：“风再大，也不至于把人吹走吧？”九月，汽车开上坝上的那天，她刚下车，一阵[3]大风就把她推得往后退了好几步。风里全是沙子，打在脸上像针扎一样。她只好用手抓住车门，心里想：同学们说的，好像也没错。

6 她向四周看去，一棵树也没有。天是灰[4]的，地是黄的，风时而从北边刮来，时而从西边刮来，好像永远不会停。带她们上来的老技术员指着远处说：“看到那棵树了吗？”周玉芬看了很久，才在一片荒地[5]中间看见了一棵孤零零[6]的落叶松[7]。

1 坝上 *bàshàng* the Bashang plateau
2 周玉芬 *Zhōu Yùfēn* Zhou Yufen
3 一阵 *yīzhèn* a gust of
4 灰 *huī* grey
5 荒地 *huāngdì* wasteland
6 孤零零 *gūlínglíng* all alone
7 落叶松 *luòyèsōng* larch

Before she left, her classmates teased her: "We hear the wind up on the plateau is strong enough to blow people away." Zhou Yufen laughed. "However strong the wind is, it can hardly blow a person away, can it?" In September, on the day the truck climbed onto the plateau, the moment she got out a great gust pushed her back several steps. The wind was full of sand, and it stung her face like needles. She had to grab the door of the truck, thinking to herself that her classmates had not been so wrong after all. 5

She looked all around: there was not a single tree. The sky was grey, the ground was yellow, and the wind blew now from the north, now from the west, as if it would never stop. The old technician who had brought them up pointed into the distance. "Do you see that tree?" Zhou Yufen looked for a long time before she made out a lone larch standing in the middle of the wasteland. 6

7 “据说去年林业部[†]的人来这里考察，走了好几天，只发现了这一棵。”技术员说，“你们想想，既然这里能长出一棵树，为什么就不能长出一百万棵？”那天晚上，周玉芬[‡]在本子上写下了第一篇日记：“今天到了坝上[§]。风很大。没有树。我们就是来种树的。”

第一年，树都死了

8 到了坝上，周玉芬才听说，这一年春天，林场[¶]已经种下了大约一千亩[*]树。可是到了秋天，据说活下来的还不到百分之五。树苗[#]是从很远的外地运来的，在路上走了好几天，根早就干了。种下去以后，又遇到坝上的大风，好几个月不下雨，小树便一棵一棵地干死了。老职工们站在地里，谁也不说话。

† 林业部 *Línyèbù* Ministry of Forestry
‡ 周玉芬 *Zhōu Yùfēn* Zhou Yufen
§ 坝上 *bàshàng* the Bashang plateau
¶ 林场 *línchǎng* forest farm
* 亩 *mǔ* mu (about 667 m²)
\# 树苗 *shùmiáo* sapling

"They say that last year, when people from the Ministry of Forestry came 7
to survey this place, they walked for days and found only this one," said the technician. "Think about it. If one tree can grow here, why not a million?" That night Zhou Yufen wrote her first diary entry in her notebook: "Arrived on the plateau today. Very windy. No trees. Planting trees is exactly what we came for."

The First Year, All the Trees Died

Only when she reached the plateau did Zhou Yufen learn that the farm had 8
already planted about a thousand mu of trees that spring. But by autumn, it was said, fewer than five per cent had survived. The saplings had been brought from far away; they were on the road for days, and their roots had long since dried out. Once planted, they met the plateau's winds and months without rain, and one after another the little trees dried up and died. The old hands stood in the fields, and no one said a word.

9 坝上[†]的冬天特别长，从十月一直到第二年四月，最冷的时候能到零下四十多度。即使到了五月，这里也常常下雪。他们住的是自己挖的“地窨子[‡]”：在地上挖一个很深的坑[§]，上面盖上木头和草，就成了一间屋子。晚上睡觉，大家都穿着棉衣、戴着帽子；早上起来，被子上常常有一层白色的冰，那是人呼出来的气冻成的。周玉芬[¶]的手冻伤了，写日记的时候，手指时而疼得连笔都拿不住。

10 第二年春天，她第一次自己动手种树。她们把小树一棵一棵放进坑里，再用脚把土踩紧。周玉芬每天都要去地里看好几次，好像怕它们跑了似的。可是一个月以后，小树的叶子开始发黄；又过了一个月，大部分都死了。那一年，林场[*]又种了一千多亩[#]，据说活下来的也不到百分之八。

† 坝上 *bàshàng* the Bashang plateau
‡ 地窨子 *dìyìnzi* dugout dwelling
§ 坑 *kēng* pit, hole
¶ 周玉芬 *Zhōu Yùfēn* Zhou Yufen
* 林场 *línchǎng* forest farm
\# 亩 *mǔ* mu (about 667 m²)

Winter on the plateau was especially long, from October right through to 9
the following April, and at its coldest it could reach more than forty degrees below zero. Even in May it often snowed here. They lived in dugouts they had made themselves: you dug a deep pit in the ground, covered it with timber and grass, and that was a room. At night everyone slept in padded coats and hats; in the morning there was often a layer of white ice on the quilts, formed from the breath people had let out. Zhou Yufen's hands were frostbitten; when she wrote her diary, her fingers at times hurt so much that she could not even hold the pen.

The next spring, she planted trees with her own hands for the first time. 10
They placed the little trees in the holes one by one, then trod the soil down firmly with their feet. Zhou Yufen went out to the fields several times a day, as if she were afraid they might run off. But a month later the little trees' needles began to turn yellow, and after another month most of them were dead. That year the farm planted over a thousand mu again, and it is said that fewer than eight per cent survived.

11 林场[†]里开始有人说：“这地方根本长不了树，再种也是白种。”还有传言说，上面打算把林场关了。秋天，有几个年轻人收拾好行李，悄悄地离开了坝上[‡]。那天晚上，周玉芬[§]怎么也睡不着。她用被子蒙住头，在里面打着手电写日记，免得吵醒同屋的姐妹。那篇日记只有一句话：“我也想走。”

12 第二天的日记更短，只有两个字：“没走。”为什么没走，她没有写。六十多年以后，孙女赵雨[¶]读到这一页的时候，也问过奶奶同样的问题。不过，那是后来的事了。那年冬天，林场到底关不关，谁也说不清。可是坝上的风照样刮着，地窨子[*]里的灯也照样每天晚上亮着。

† 林场 *línchǎng* forest farm
‡ 坝上 *bàshàng* the Bashang plateau
§ 周玉芬 *Zhōu Yùfēn* Zhou Yufen
¶ 赵雨 *Zhào Yǔ* Zhao Yu
* 地窨子 *dìyìnzi* dugout dwelling

People on the farm began to say, "Trees simply can't grow here. Planting more is a waste of effort." There was even a rumour that the authorities meant to close the farm. In the autumn a few young people packed their bags and quietly left the plateau. That night Zhou Yufen could not get to sleep, no matter what. She pulled the quilt over her head and wrote her diary underneath it by torchlight, so as not to wake the girls she shared the room with. That entry had just one line: "I want to leave too." 11

The next day's entry was even shorter, just two characters: "Didn't leave." Why she didn't leave, she did not write. More than sixty years later, when her granddaughter Zhao Yu read this page, she asked her grandmother the very same question. But that came later. That winter, nobody could say for sure whether the farm would be closed or not. But the wind on the plateau blew just as before, and the lamps in the dugouts still came on every evening as usual. 12

马蹄坑[1]

13 失败了两次，林场[2]的技术员们开始认真地找原因。他们发现，外地的树苗[3]不适应坝上[4]的气候，路上又丢了水分，最好的办法是在坝上自己培育树苗。他们还改进了种树的方法：坑[5]挖多深，根怎么放，土踩几下，每一步都重新试过。周玉芬[6]是学林业的，被派去培育树苗。那时候什么条件都没有，她们凭着几本旧书和一次又一次的失败，总算在坝上养出了第一批又矮又壮[7]的小落叶松[8]。

14 一九六四年春天，林场决定再试一次。地方选在一个叫“马蹄坑”的山谷里，据说是因为那块地的样子像一个马蹄[9]印。这一次，林场的领导亲自带着大家，在山坡上住了下来。四月的坝上还很冷，夜里地上结着冰，他们就睡在帐篷[10]里。

1 马蹄坑 *Mǎtíkēng* Horseshoe Hollow
2 林场 *línchǎng* forest farm
3 树苗 *shùmiáo* sapling
4 坝上 *bàshàng* the Bashang plateau
5 坑 *kēng* pit, hole
6 周玉芬 *Zhōu Yùfēn* Zhou Yufen
7 壮 *zhuàng* sturdy
8 落叶松 *luòyèsōng* larch
9 蹄 *tí* hoof
10 帐篷 *zhàngpeng* tent

Horseshoe Hollow

After failing twice, the farm's technicians set about seriously looking for 13
the causes. They found that saplings from elsewhere could not adapt to the plateau's climate and lost moisture on the road besides; the best solution was to raise saplings on the plateau themselves. They also improved the way they planted: how deep to dig the hole, how to set the roots, how many times to tread the soil, every step was tried again from scratch. Since Zhou Yufen had studied forestry, she was sent to raise saplings. They had nothing to work with in those days, but relying on a few old books and failure after failure, they finally grew their first batch of short, sturdy little larches on the plateau.

In the spring of 1964 the farm decided to try once more. The place cho- 14
sen was a hollow called Horseshoe Hollow, supposedly because the land was shaped like a hoofprint. This time the farm's leaders came along in person and settled on the hillside with everyone. April on the plateau was still cold, the ground froze at night, and they slept in tents.

14 平地上可以用机器，机器上不去的山坡，就全凭一双双手。

15 在马蹄坑[1]，周玉芬[2]认识了一个从东北来的小伙子，叫赵长山[3]。他话不多，力气很大，一天能挖别人两倍的坑[4]。有一次，大风把周玉芬的帽子吹跑了，他追了很远才追回来。他把帽子递给她，只说了一句："戴好，免得冻坏耳朵。"那是他对她说的第一句话。两年以后，他们结了婚。

16 一个月以后，周玉芬跟着技术员一棵一棵地检查，小树都长出了新的绿叶。到了秋天，结果出来了：马蹄坑种下的五百多亩[5]落叶松[6]，活下来的超过了百分之九十。消息传回林场[7]，大家高兴得又叫又跳，有的人哭了。那天的日记里，周玉芬写道："活了！树总算活了！"

1 马蹄坑 *Mǎtíkēng* Horseshoe Hollow
2 周玉芬 *Zhōu Yùfēn* Zhou Yufen
3 赵长山 *Zhào Chángshān* Zhao Changshan
4 坑 *kēng* pit, hole
5 亩 *mǔ* mu (about 667 m²)
6 落叶松 *luòyèsōng* larch
7 林场 *línchǎng* forest farm

On flat ground they could use machines, but on the slopes the machines could not climb, it all came down to human hands. 14

At Horseshoe Hollow Zhou Yufen got to know a young man from the Northeast called Zhao Changshan. He did not say much, but he was very strong and could dig twice as many holes in a day as anyone else. Once the wind blew Zhou Yufen's hat away, and he chased it a long way before he got it back. He handed her the hat and said only, "Put it on properly, so you don't freeze your ears." Those were the first words he ever said to her. Two years later they were married. 15

A month later Zhou Yufen went along with the technicians checking tree after tree, and every little tree had put out new green needles. In the autumn the results came in: of the more than five hundred mu of larch planted at Horseshoe Hollow, over ninety per cent had survived. When the news reached the farm, people jumped and shouted for joy, and some wept. In her diary that day Zhou Yufen wrote: "They're alive! The trees are finally alive!" 16

16 也正是从马蹄坑[1]开始，塞罕坝[2]人才真正相信：这里能长出一片森林。

望火楼[3]

17 周玉芬[4]和赵长山[5]的儿子出生在一九六七年，名字叫赵建林[6]，意思是“建设森林”。建林[7]是跟着林子一起长大的。他小的时候，林子还没有他高；等他长成了小伙子，林子已经比房子还高了。他十岁那年秋天，坝上[8]下了一场少见[9]的冻雨[10]，厚厚的冰把大片[11]的树压弯、压断，据说受灾的林子有五十多万亩[12]。建林记得，父亲站在断了的树中间，很久都没有说话。第二年春天，父亲和母亲又扛[13]着铁锹[14]上了山。

1 马蹄坑 *Mǎtíkēng* Horseshoe Hollow
2 塞罕坝 *Sàihǎnbà* Saihanba, forest in Hebei
3 望火楼 *wànghuǒlóu* fire lookout tower
4 周玉芬 *Zhōu Yùfēn* Zhou Yufen
5 赵长山 *Zhào Chángshān* Zhao Changshan
6 赵建林 *Zhào Jiànlín* Zhao Jianlin
7 建林 *Jiànlín* Jianlin
8 坝上 *bàshàng* the Bashang plateau
9 少见 *shǎojiàn* rare
10 冻雨 *dòngyǔ* freezing rain
11 大片 *dàpiàn* large stretch
12 亩 *mǔ* mu (about 667 m²)
13 扛 *káng* to shoulder
14 铁锹 *tiěqiāo* spade

And it was from Horseshoe Hollow on that the people of Saihanba truly came to believe that a forest could grow here. 16

The Fire Tower

Zhou Yufen and Zhao Changshan's son was born in 1967. They named him Zhao Jianlin, meaning "build a forest". Jianlin grew up together with the woods. When he was small, the trees were not as tall as he was; by the time he was a young man, they were taller than the houses. In the autumn of the year he turned ten, a rare freezing rain fell on the plateau. Thick ice bent and snapped great stretches of trees; it is said that more than half a million mu of forest were damaged. Jianlin remembered his father standing among the broken trees, not saying a word for a long time. The next spring, his father and mother shouldered their spades and went back up the hills. 17

18 二十二岁那年，建林[†]主动要求去守望火楼[‡]。望火楼建在林场[§]很高的一个山头上，四面都是窗户，站在里面，几十公里内的林子都看得清清楚楚。每年的春天和秋天最容易着火：草干了，风又大，一个烟头、一道闪电，都可能烧掉几十年的辛苦。守楼的人每隔十五分钟就要把四周看一遍，再打电话向林场报告一次，免得错过任何一点烟。

19 楼里只有一张床、一个炉子[¶]和一部电话。吃的水要从山下背上来，蔬菜一个星期才送一次。最难受的不是冷，而是安静。有时候一整天，除了电话里那几句“一切正常”，他一句话也说不上。母亲上山来看他，担心地问：“一个人在山上一住就是半年，不至于孤独得受不了吧？”建林笑了笑：“哪儿会呢？我有树陪着。”

20 他慢慢学会了跟林子打交道。

† 建林 *Jiànlín* Jianlin
‡ 望火楼 *wànghuǒlóu* fire lookout tower
§ 林场 *línchǎng* forest farm
¶ 炉子 *lúzi* stove

At twenty-two, Jianlin volunteered to keep watch in a fire tower. The tower 18
stood on one of the highest hilltops on the farm, with windows on all four sides; from inside you could see the woods clearly for dozens of kilometres around. Spring and autumn were when fires broke out most easily: the grass was dry and the wind was strong, and a single cigarette end or a flash of lightning could burn away decades of hard work. Whoever kept the tower had to scan all around every fifteen minutes and then phone in a report to the farm, so as not to miss the slightest trace of smoke.

In the tower there was only a bed, a stove and a telephone. Drinking water 19
had to be carried up on his back from the foot of the hill, and vegetables were delivered only once a week. The hardest part was not the cold but the silence. Some days, apart from those few words of "all normal" on the phone, he did not get to say a single thing all day. His mother came up the hill to see him and asked anxiously, "Living alone up here for half a year at a stretch, it's not so lonely you can't bear it, is it?" Jianlin smiled. "How could it be? I have the trees to keep me company."

Little by little he learned how to get along with the woods. 20

20 白天，他时而拿起望远镜[†]看看远处，时而在本子上记下风向和温度；晚上，听着风吹过松树的声音，他就知道第二天是什么天气。他也学会了分清烟和云：云是散的，会慢慢移动；烟是从地面升起来的，下面细，上面粗，颜色也更灰[‡]。

21 有一年秋天的下午，天上突然打了一个雷。过了几分钟，他在东南方向看见了一条细细的灰线。他马上拿起电话："东南方向，大约十二公里，有烟！"半个小时以后，救火的人赶到了。因为发现得早，火只烧了不到一亩[§]地，就被扑灭了。那天夜里，建林[¶]一夜没睡。他想，要是晚看见二十分钟，烧掉的也许就是母亲那一代人种了几十年的树。

22 他在望火楼[*]上一守就是二十多年。女儿赵雨[#]小的时候，每年暑假都会上山陪他住几天。

† 望远镜 *wàngyuǎnjìng* binoculars
‡ 灰 *huī* grey
§ 亩 *mǔ* mu (about 667 m²)
¶ 建林 *Jiànlín* Jianlin
* 望火楼 *wànghuǒlóu* fire lookout tower
\# 赵雨 *Zhào Yǔ* Zhao Yu

By day he would now raise the binoculars to look into the distance, now note down the wind direction and temperature in his notebook; at night, listening to the wind in the pines, he could tell what the weather would be the next day. He also learned to tell smoke from cloud: cloud is loose and drifts slowly; smoke rises from the ground, thin at the bottom and broad at the top, and greyer in colour. 20

One autumn afternoon there was a sudden clap of thunder. A few minutes later he saw a thin grey line to the southeast. He grabbed the phone at once. "Southeast, about twelve kilometres, smoke!" Half an hour later the firefighters arrived. Because it had been spotted early, the fire burned less than a mu of land before it was put out. That night Jianlin did not sleep at all. He thought that if he had seen it twenty minutes later, what burned might have been the trees his mother's generation had spent decades planting. 21

Once he took up his post in the tower, he stayed for more than twenty years. When his daughter Zhao Yu was little, she came up the hill every summer holiday to stay with him for a few days. 22

22 他教她看风、看云、看烟，告诉她：“守林子，靠的是眼睛。”那时候他没有想到，很多年以后，女儿会用另一种“眼睛”守着这片林子。

奶奶的日记

23 赵雨[†]在北京读完了林业大学的硕士，学的是遥感[‡]，就是用卫星和飞机上的“眼睛”去看森林。毕业的时候，北京的一家公司给她开了很高的工资，同学们都劝她留下。她却回到了塞罕坝[§]，在林场[¶]做数据工作。父亲什么也没说，只是那天晚上多喝了两杯酒。奶奶倒是很直接：“回来好。林子需要年轻人。”

24 那年冬天，奶奶已经八十多岁了，眼睛越来越不好，看报纸都得用放大镜。

† 赵雨 *Zhào Yǔ* Zhao Yu
‡ 遥感 *yáogǎn* remote sensing
§ 塞罕坝 *Sàihǎnbà* Saihanba, forest in Hebei
¶ 林场 *línchǎng* forest farm

He taught her to read the wind, the clouds and the smoke, and told her, 22
"Guarding a forest depends on your eyes." He never imagined then that years later his daughter would guard these woods with a different kind of eye.

Grandmother's Diary

Zhao Yu did a master's degree at a forestry university in Beijing. She studied 23
remote sensing, which means looking at forests with "eyes" on satellites and aircraft. When she graduated, a company in Beijing offered her a high salary, and her classmates all urged her to stay. But she went back to Saihanba to do data work for the forest farm. Her father said nothing; he just had a couple more drinks than usual that night. Her grandmother, for her part, was blunt: "Good that you're back. The forest needs young people."

That winter her grandmother was over eighty, and her eyes were getting 24
worse and worse; she needed a magnifying glass even to read the paper.

24 有一天，她从柜子最下面拿出一个用布包着的旧本子，交给赵雨[†]：“我看不清了，你念给我听吧。”赵雨打开一看，是一本日记，从一九六二年写起，前前后后写了好几年。从那以后，每天晚饭后，赵雨便坐在奶奶床边，念上几页。

25 奶奶的字时而整整齐齐，时而乱得几乎认不出来。赵雨念到后来才明白：字乱的那几页，大多是冬天写的，那时候手冻得拿不住笔。日记里很少出现“苦”字，写得最多的是天气、树，还有“今天种了多少”。有一页只写着：“零下三十八度。种了四百棵。小李[‡]的脚冻坏了。”赵雨念着念着，声音就低了下去。

26 念到“我也想走”和“没走”那两页，赵雨停了下来。“奶奶，那天您为什么没走？”奶奶闭着眼睛想了很久。“第二天早上，我想去地里再看最后一次。”她慢慢地说，“一大片树都死了，只有一棵还绿着，小小的，还没有我的手高。

† 赵雨 *Zhào Yǔ* Zhao Yu

‡ 小李 *Xiǎo Lǐ* Xiao Li

One day she took an old cloth-wrapped notebook from the bottom of the cupboard and handed it to Zhao Yu. "I can't see well any more. Read it to me." Zhao Yu opened it and saw that it was a diary, begun in 1962 and kept, on and off, for several years. From then on, every evening after dinner, Zhao Yu would sit by her grandmother's bed and read a few pages. 24

Her grandmother's handwriting was sometimes perfectly neat, and sometimes such a scrawl it could hardly be read. Only as she read on did Zhao Yu understand that the messy pages had mostly been written in winter, when her grandmother's hands were too frozen to hold the pen. The word "hardship" hardly ever appeared; what she wrote about most was the weather, the trees, and "how many we planted today". One page said only: "Thirty-eight below. Planted four hundred. Xiao Li's feet are frostbitten." As Zhao Yu read on, her voice grew quieter and quieter. 25

When she reached the two pages "I want to leave too" and "Didn't leave", Zhao Yu stopped. "Grandma, why didn't you leave that day?" Her grandmother closed her eyes and thought for a long time. "The next morning I wanted to go and look at the fields one last time," she said slowly. "A whole stretch of trees was dead, and only one was still green, a tiny thing, no taller than my hand. 26

26 我在那儿蹲了半天，心想：我要是走了，谁来照顾它呢？”

27 “就因为一棵树？”“一棵树嘛，当然不至于让一个人留下来。”奶奶笑了，“可是你看见了那一棵，就会想：还会有第二棵、第三棵。”赵雨[†]忽然明白了：这片森林不是凭什么奇迹[‡]长出来的，而是凭一个又一个这样的早上。即使只剩一棵绿的，也有人愿意再种一年。

无人机[§]

28 在林场[¶]，赵雨每天的工作离不开无人机和卫星照片。无人机从林子上面飞过，能拍下每一棵树的样子；电脑再把几千张照片拼在一起，算出每棵树有多高、长得快不快、有没有生病。

† 赵雨 *Zhào Yǔ* Zhao Yu
‡ 奇迹 *qíjì* miracle
§ 无人机 *wúrénjī* drone
¶ 林场 *línchǎng* forest farm

I squatted there for ages, thinking: if I leave, who will look after it?" 26

"Just because of one tree?" "One tree, of course, wouldn't be enough to 27
make anyone stay." Her grandmother smiled. "But once you've seen that one, you start to think there will be a second, and a third." Zhao Yu suddenly understood: this forest had not grown out of some miracle, but out of one such morning after another. Even when there was only one green tree left, someone was willing to plant for another year.

Drones

At the farm, Zhao Yu's daily work revolved around drones and satellite im- 28
ages. A drone flying over the woods can photograph every tree; the computer then pieces thousands of images together and works out how tall each tree is, whether it is growing fast, and whether it is sick.

28 以前要很多人走好几个月才能做完的调查，现在几天就能完成。即使是在夜里，带热成像[1]镜头的无人机[2]也能发现地面上不正常的热点。

29 父亲那时已经从望火楼[3]上下来了，可他对女儿的这些“新眼睛”不太放心。“机器能看见烟？”他说，“我在楼上看了二十多年，是烟是云，看一下就知道。机器分得清吗？”“分得清，还能看见您看不见的东西。”赵雨[4]把电脑转过去，“您看，哪儿温度高，哪儿就是红的。”父亲看了半天，只说了一句：“花样[5]倒不少。”

30 那年夏天，赵雨在数据里发现了一个奇怪的地方：林场[6]北边有一小片落叶松[7]，颜色比旁边的浅一点。从地上看，那些树好好的，什么问题也没有；可是无人机每飞一次，那一片的颜色就比上次更浅一点。她和父亲一起开车去看。

1 热成像 *rèchéngxiàng* thermal imaging
2 无人机 *wúrénjī* drone
3 望火楼 *wànghuǒlóu* fire lookout tower
4 赵雨 *Zhào Yǔ* Zhao Yu
5 花样 *huāyàng* fancy tricks
6 林场 *línchǎng* forest farm
7 落叶松 *luòyèsōng* larch

Surveys that once took many people months of walking can now be finished in a few days. Even at night, a drone with a thermal imaging camera can pick out unusual hot spots on the ground. 28

By then her father had come down from the fire tower, but he did not quite trust his daughter's "new eyes". "A machine can see smoke?" he said. "I watched from that tower for over twenty years; smoke or cloud, I could tell at a glance. Can a machine tell them apart?" "It can, and it can see things you can't." Zhao Yu turned the computer round. "Look: wherever it's hotter, it shows up red." Her father looked for a long time and said only, "Plenty of fancy tricks, I'll give it that." 29

That summer Zhao Yu found something odd in the data: a small patch of larch in the north of the farm was slightly paler than the trees around it. From the ground the trees looked perfectly fine; but each time the drone flew over, the patch was a little paler than the time before. She and her father drove out to take a look. 30

30 父亲拿着放大镜，在树叶中间找了半天，终于找到了几条小小的虫子。“是松毛虫[†]。”他的脸色变了，“再过一个月，这片林子就完了。”林场[‡]马上派人处理了那一片树，免得虫子传到别的林子里去。

31 回来的路上，父亲一直看着窗外。过了很久，他才说话：“有了这些机器，守望火楼[§]的人，不至于就没用了吧？”赵雨[¶]想了想，说：“爸，机器只能告诉我们哪儿有问题。可是走过去看、想办法的，还是人。今天找到虫子的，不也是您吗？”父亲没有回答，可赵雨看见他笑了。那是她第一次觉得，父亲和她守着的是同一片林子。

32 二〇一七年，塞罕坝[*]的建设者们获得了联合国环境规划署[#]的“地球卫士奖”。

† 松毛虫 *sōngmáochóng* pine caterpillar
‡ 林场 *línchǎng* forest farm
§ 望火楼 *wànghuǒlóu* fire lookout tower
¶ 赵雨 *Zhào Yǔ* Zhao Yu
* 塞罕坝 *Sàihǎnbà* Saihanba, forest in Hebei
联合国环境规划署 *Liánhéguó Huánjìng Guīhuàshǔ* UN Environment Programme

Her father searched among the needles with a magnifying glass for a long time and at last found a few tiny caterpillars. "Pine caterpillars." His face changed. "Another month and this stand would be finished." The farm sent people at once to treat that patch, so that the caterpillars would not spread to other stands. 30

All the way back, her father kept looking out of the window. After a long while he finally spoke. "With all these machines, the people who keep the fire towers aren't going to be useless now, are they?" Zhao Yu thought for a moment. "Dad, machines can only tell us where there's a problem. It's still people who go and look and work out what to do. Wasn't it you who found the caterpillars today?" Her father did not answer, but Zhao Yu saw him smile. It was the first time she felt that she and her father were guarding the same forest. 31

In 2017 the people who built Saihanba received the United Nations Environment Programme's "Champions of the Earth" award. 32

32 这时，塞罕坝[1]的森林已经有一百多万亩[2]，据说是世界上面积最大的人工林[3]之一。森林覆盖率[4]从当初的百分之十一左右，增加到了百分之八十以上。这片林子为北京和天津[5]保护着水源[6]，也挡住了北边吹来的风沙[7]。新闻里的这些数字，赵雨[8]都很熟悉。可是她知道，每一个数字后面，都有一本像奶奶那样的日记。

一片林子

33 十月，坝上[9]的落叶松[10]全黄了。赵雨开着车，带奶奶和父亲去马蹄坑[11]。出门前，父亲给奶奶多穿了一件厚衣服，免得她在山上着凉。一路上，奶奶一直看着窗外。

1 塞罕坝 *Sàihǎnbà* Saihanba, forest in Hebei
2 亩 *mǔ* mu (about 667 m²)
3 人工林 *réngōnglín* planted forest
4 覆盖率 *fùgàilǜ* forest cover rate
5 天津 *Tiānjīn* Tianjin
6 水源 *shuǐyuán* water source
7 风沙 *fēngshā* wind-blown sand
8 赵雨 *Zhào Yǔ* Zhao Yu
9 坝上 *bàshàng* the Bashang plateau
10 落叶松 *luòyèsōng* larch
11 马蹄坑 *Mǎtíkēng* Horseshoe Hollow

By then Saihanba's forest covered more than a million mu, and it is said to be one of the largest planted forests in the world. Forest cover rose from around eleven per cent at the start to more than eighty per cent. These woods protect water sources for Beijing and Tianjin, and hold back the wind-blown sand from the north. Zhao Yu knew all these figures from the news by heart. But she knew that behind every figure there was a diary like her grandmother's. 32

A Forest

In October the larches on the plateau turned completely golden. Zhao Yu drove her grandmother and father to Horseshoe Hollow. Before they set out, her father put an extra thick layer on Grandmother so that she wouldn't catch cold up in the hills. All the way there, Grandmother kept looking out of the window. 33

33 六十多年前，这条路的两边只有沙子和野草；现在，车在林子里开了一个多小时，还没有开出去。

34 马蹄坑[†]的落叶松[‡]已经有二十多米高了，树干粗得两只手都抱不过来。奶奶扶着赵雨[§]的手，慢慢走到一棵树下，用手摸着树皮，摸了很久。

“这一片，是我们种的。”她说，“那时候它们还没有筷子高。”父亲站在后面，轻轻地说：

“妈，您总算又回来看它们了。”

35 赵雨让无人机[¶]飞了起来。它越飞越高，屏幕上的林子也越来越大：近处是金黄的落叶松，远处是深绿的樟子松[*]，中间还有几个小湖，在太阳下闪着光。奶奶把屏幕拿到眼前，看了很久。

“原来是这个样子。”她说，“种了一辈子树，我还是第一次从天上看它。”

† 马蹄坑 *Mǎtíkēng* Horseshoe Hollow
‡ 落叶松 *luòyèsōng* larch
§ 赵雨 *Zhào Yǔ* Zhao Yu
¶ 无人机 *wúrénjī* drone
* 樟子松 *zhāngzǐsōng* Mongolian Scots pine

More than sixty years ago, there had been nothing but sand and wild grass on either side of this road; now they had driven through the woods for over an hour and still had not come out of them. 33

The larches at Horseshoe Hollow were now more than twenty metres tall, their trunks too thick to get both arms around. Leaning on Zhao Yu's hand, Grandmother walked slowly up to a tree and stroked its bark for a long time. "We planted this stand," she said. "Back then they weren't as tall as a chopstick." Her father, standing behind, said softly, "Mum, you've finally come back to see them again." 34

Zhao Yu sent the drone up. The higher it flew, the larger the forest on the screen grew: golden larches close by, dark green Mongolian pines in the distance, and among them a few small lakes glinting in the sun. Grandmother held the screen up close to her eyes and looked for a long time. "So that's what it looks like," she said. "I spent my whole life planting trees, and this is the first time I've seen them from the sky." 35

36 那天晚上，赵雨†翻开奶奶的日记，发现最后几页还是空的。奶奶说：“你来写吧。”赵雨想了想，拿起笔，写下：“今天去了马蹄坑‡。风很大。到处都是树。”她看了看奶奶，又加了一句：“我们还会接着种。”窗外，风从林子里吹过，声音像大海一样。

† 赵雨 *Zhào Yǔ* Zhao Yu

‡ 马蹄坑 *Mǎtíkēng* Horseshoe Hollow

That evening Zhao Yu opened her grandmother's diary and found that the last few pages were still blank. Her grandmother said, "You write in it." Zhao Yu thought for a moment, picked up the pen and wrote: "Went to Horseshoe Hollow today. Very windy. Trees everywhere." She looked at her grandmother and added one more line: "We will keep on planting." Outside the window the wind blew through the woods with a sound like the sea. 36